PropRent

Property Rental Management & Tenant Portal Platform

Chapter 1: Project Planning & Management

Version 1.0

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Baseline	Requirements Gathering Specification v1.0

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Document Control

Item	Details
Chapter	Chapter 1: Project Planning & Management
Version	1.0
Author / Owner	Asmaa Hany (System Lead)
Contributors	Ahmed Fouda, Saad Walid, Yara Mokhtar, Manar Gomaa, Hager Saad
DEPI Submission Deadline	16 October 2026
Baseline Reference	PropRent Requirements Gathering Specification v1.0 (108 functional requirements, 7 epics, 4 NFR categories + maintainability)
Change Control	Changes to scope or schedule follow the procedure defined in Section 1.2.2

1.1 Project Proposal

1.1.1 Executive Summary & Overview

Business context. Residential rental management in many markets, including Egypt's growing formal rental sector, still relies on a patchwork of spreadsheets, messaging applications, paper leases, phone calls and manual bank transfers. A single property manager may coordinate dozens of units, each with its own lease terms, payment schedule, maintenance history and communication thread. Owners, tenants and contractors each hold a partial view of the same reality.

PropRent is a web-based Property Rental Management & Tenant Portal Platform that consolidates the full rental lifecycle (listing, application, lease, billing, payment, maintenance, communication and reporting) into a single, role-aware system serving tenants, property managers, owners, contractors and platform administrators.

Problems addressed.

#	Problem	Observable Consequence	PropRent Response
P1	**Fragmentation of rental management:** data and workflows are split across spreadsheets, email, chat and paper.	Duplicated data entry, lost documents, no audit trail, high administrative load per unit.	One system of record for properties, units, leases, documents and tickets, with role-based access (Epics AUTH, PROP, LEASE).
P2	**Tenant communication delays:** repairs are reported by phone, notices are informal and replies are slow.	Unresolved defects, tenant dissatisfaction, missed legal notices, no communication archive.	Structured maintenance tickets with media, SLA tracking, threaded messaging, broadcast announcements and multi-channel notifications (Epics MAINT, COMM).
P3	**Financial opacity:** rent status, balances and owner income are unclear or reconciled manually.	Missed due dates, disputed payments, delayed owner statements, unreconciled cash.	Automated invoicing, gateway-based online payment, append-only double-entry ledger, and real-time dashboards and statements (Epics FIN, RPT).

Value proposition by stakeholder.

- **Tenants:** one place to pay rent, view the lease and receipts, report repairs and track their status.
- **Property managers:** a centralised portfolio view, automated billing and reminders, structured work orders.
- **Owners:** transparent income, occupancy and expense reporting, with approval control above spending thresholds.
- **Administrators:** secure user and role administration, configuration without code changes, and audit logs.

Proposed solution at a glance. A modular, API-first platform (modular monolith with module boundaries aligned to the seven epics, per NFR-MNT-01), integrating a PCI-DSS Level 1 payment gateway, an e-signature provider and email/SMS notification providers behind provider-agnostic interfaces (NFR-MNT-12), and deployed in containers on a cloud provider (NFR-MNT-09).

1.1.2 Project Objectives (SMART)

ID	Objective	Specific	Measurable	Achievable	Relevant	Time-bound
OBJ-1	**Core business value:** deliver an integrated platform covering the full rental lifecycle.	All 7 epics (AUTH, PROP, LEASE, FIN, MAINT, COMM, RPT) delivered as working modules.	100% of the 63 Must-Have requirements implemented and demonstrated; at least 80% of the 27 Should-Have requirements.	Scope is prioritised with MoSCoW; Must-Haves are built first.	Resolves fragmentation (P1).	Implementation complete by 30 Nov 2026.
OBJ-2	**Rent billing automation:** eliminate manual invoicing and rent chasing.	Scheduler generates invoices idempotently (FR-FIN-01), applies late fees (FR-FIN-10), and sends reminders (FR-COMM-08).	100% of active-lease invoices generated automatically without manual action; zero duplicate invoices; reminders sent before due date, on due date and after grace period.	Uses a standard job scheduler and queue workers (NFR-PERF-11).	Reduces manager workload and missed payments (P3).	Demonstrable end-to-end in staging by 20 Nov 2026.
OBJ-3	**End-to-end maintenance SLA tracking:** make every repair visible from submission to verified closure.	Ticket lifecycle, priority-based SLA targets, contractor assignment, escalation on breach (FR-MAINT-01 to -14).	100% of tickets carry a unique ID and time-stamped history; 100% of SLA breaches escalate on the manager dashboard; ticket flow covered by automated end-to-end tests.	Workflow and notification components reuse the COMM module.	Resolves communication delays (P2).	Demonstrable in staging by 20 Nov 2026.
OBJ-4	**Audit-proof financial records:** guarantee traceable, tamper-evident money movement.	Append-only double-entry ledger per property and lease (FR-FIN-14); signed, deduplicated webhooks (FR-FIN-08); audit log (FR-AUTH-13).	Zero unexplained variances in daily gateway-to-ledger reconciliation (NFR-REL-07); at least 90% unit-test coverage on the financial module (NFR-MNT-04).	Gateway sandbox and test-mode support deterministic reconciliation testing.	Resolves financial opacity (P3) and is critical for owner trust.	Verified in final test report by 2 Dec 2026.
OBJ-5	**Quality and delivery discipline:** meet DEPI milestones with a quality-gated release.	CI/CD pipeline with build, tests and security scans on every commit (NFR-MNT-05).	All 4 DEPI milestones met on time; zero unresolved critical/high vulnerabilities at release (NFR-SEC-09); at most 1 critical defect at release (NFR-REL-10).	Pipeline is established in Sprint 0 and enforced on every merge.	Ensures a defensible, demonstrable graduation deliverable.	Final presentation 4 Dec 2026.

1.1.3 Project Scope

1.1.3.1 In-Scope

The project scope is the baseline defined in the Requirements Gathering Specification v1.0. The seven functional epics, with their MoSCoW distribution, are:

Epic	Module Code	Must	Should	Could	Won't	Total
Epic 1: Authentication, RBAC & Profile Management	AUTH	11	2	1	1	15
Epic 2: Property & Unit Listing Management	PROP	8	4	2	1	15
Epic 3: Tenant Onboarding, Lease Agreements & Document Management	LEASE	9	4	1	1	15
Epic 4: Financial Engine, Rent Billing & Payment Processing	FIN	11	5	2	1	19
Epic 5: Maintenance & Work Order Ticketing System	MAINT	9	5	2	1	17
Epic 6: Notifications, Messaging & Communication Hub	COMM	7	3	2	1	13
Epic 7: Reporting, Analytics & Manager Dashboard	RPT	8	4	1	1	14
Total		**63**	**27**	**11**	**7**	**108**

Committed scope: all 63 Must-Have and 27 Should-Have requirements (90 requirements), plus the non-functional requirements in the specification (performance, security, usability, reliability, maintainability).

Scope handling rules.

- **Must-Have (63):** committed baseline; delivery is mandatory for the final demonstration.
- **Should-Have (27):** committed, delivered in priority order after the related Must-Haves within each sprint. If capacity is exhausted, lowest-value Should-Haves are re-prioritised through change control, never silently dropped.
- **Could-Have (11):** not committed; pulled only if a sprint finishes ahead of plan (e.g., social login, bulk CSV import, real-time chat via WebSockets).
- **Won't-Have (7):** formally excluded (Section 1.1.3.2).

1.1.3.2 Out-of-Scope (Deferred Features)

The following seven Won't-Have requirements were identified during requirements gathering and are explicitly deferred to a future release:

Req. ID	Deferred Feature	Rationale for Deferral
FR-AUTH-15	Enterprise SAML 2.0 SSO for property management companies	Enterprise-customer need; high integration and testing effort; no impact on core MVP workflows.
FR-PROP-15	Syndication of listings to third-party property portals	Depends on external portal agreements and APIs outside project control.
FR-LEASE-15	Automated tenant screening (credit and background reports)	Requires regulated third-party providers, applicant consent flows and legal review.
FR-FIN-19	Native accounting-ERP synchronisation	Complex per-vendor mapping; mitigated in part by the Could-Have CSV accounting export (FR-FIN-18).
FR-MAINT-17	IoT sensor-triggered tickets	Requires smart-building hardware and device integration unavailable in the project environment.

Req. ID	Deferred Feature	Rationale for Deferral
FR-COMM-13	Automatic message translation	Needs a translation service and quality assurance; manual language preference (FR-AUTH-12) is retained.
FR-RPT-14	AI/ML predictive analytics (churn, rent pricing, maintenance failure forecasting)	Needs substantial historical data and ML engineering beyond the project's timeframe.

Additional exclusions. Production-grade legal certification (e.g., formal PCI-DSS assessment beyond SAQ A scope), real-money payment processing, and live regulatory filings are excluded; the project operates in sandbox / test-mode environments.

1.1.3.3 Assumptions & Constraints

Type	Statement
Assumption	Third-party providers (payment gateway, e-signature, SMS, email) offer free sandbox or test-mode accounts for the duration of the project.
Assumption	Team members are available for the planned weekly capacity; academic and DEPI sessions are accounted for in sprint planning.
Constraint	Fixed DEPI deadlines (Section 1.2.2) cannot move; scope is the flexible variable.
Constraint	Budget is limited to student/free-tier cloud and sandbox resources.
Constraint	The team of six covers all roles (development, design, QA, DevOps and documentation).

1.2 Project Plan & Schedule

1.2.1 Methodology: Agile-Scrum Adapted to DEPI Deliverables

PropRent follows Scrum, adapted so that a fixed set of academic and DEPI documentation milestones runs in parallel with iterative software delivery ("dual-track" delivery).

Element	Adaptation for PropRent
Sprint length	2 weeks for Sprints 0 to 2; a 1-week Sprint 3 and a 4-day stabilisation sprint at the end, aligned to DEPI deadlines.
Product Owner	Ahmed Fouda (Application Lead), who owns the product backlog and prioritises by MoSCoW.
Scrum Master / Delivery	Yara Mokhtar (Delivery Lead), who facilitates ceremonies, removes blockers and tracks delivery metrics.
Development Team	All six members; epics are assigned according to Section 1.3.
Backlog	Requirements FR-IDs are the backlog items; each is traceable to user stories (US-01 to US-11) and use cases (UC-01 to UC-03).
Ceremonies	Sprint Planning (start of sprint), Daily 15-minute stand-up (asynchronous when needed), Sprint Review/Demo with supervisor feedback, Retrospective.
Definition of Ready	Story has acceptance criteria (Given-When-Then), FR traceability, UI design reference and estimated effort.
Definition of Done	Code reviewed and merged; unit tests pass (coverage gates met); security and dependency scans clean; API documented in OpenAPI; deployed to staging; acceptance criteria demonstrated.
Tooling	Git (feature-branch workflow, protected main), issue tracker/Kanban board, CI/CD pipeline, Figma, Postman/OpenAPI, shared documentation repository.
Tracking	Burndown/burn-up per sprint, velocity, and a milestone checklist mapped to DEPI deadlines.

Dual-track principle. Documentation phases (planning, literature review, requirements, analysis and design) are treated as time-boxed "documentation sprints" that feed the software sprints. Design outputs approved on 6 Nov freeze the architecture baseline for later sprints; Sprint 0 and Sprint 1 begin foundational implementation in parallel to avoid idle development time.

1.2.2 Milestones & Deliverables (Mapped to Official DEPI Deadlines)

#	DEPI Item	DEPI Deadline	Key Deliverables	Owner (Lead)	Acceptance Criteria
M1	Project Planning & Management	16 Oct 2026	This chapter: proposal, schedule, RACI, risk register, KPIs.	Asmaa Hany	Supervisor review; all five sections complete; consistent with requirements baseline.
M1	Literature Review	16 Oct 2026	Review of existing property-management platforms, payment/e-signature approaches, and relevant standards.	Asmaa Hany	Comparative analysis with citations; identified gaps justify PropRent.
M1	Requirements Gathering	16 Oct 2026	Requirements Gathering Specification v1.0 (stakeholders, 11 user stories, 3 use cases, 108 FRs, NFRs).	Asmaa Hany	Completed; baselined as v1.0; traceable IDs.
M2	System Analysis & Design	6 Nov 2026	Architecture, ERD/data dictionary, API contracts, UML (use case, sequence, class, state), UI/UX wireframes and prototypes, deployment design, ledger design.	Asmaa Hany with Saad Walid	Design review sign-off; every Must-Have requirement mapped to a design element.
M3	Implementation (Source Code & Execution)	30 Nov 2026	Working system in staging: all 7 epics, CI/CD pipeline, OpenAPI docs, deployment guide, seed/demo data.	Ahmed Fouda with Hager Saad	63/63 Must-Haves and target Should-Haves running on staging; code freeze on 30 Nov.
M4	Final Presentation, Testing & Reports	4 Dec 2026	Test report (unit, integration, E2E, performance, security), traceability matrix, final documentation, presentation and live demo.	Yara Mokhtar and Manar Gomaa	KPI evidence collected; demo script rehearsed; final documents submitted.

Change control (summary). Any change to the requirements baseline or milestone dates requires: (1) a written change request stating the reason and impact on scope, schedule and risk; (2) review by the Application Lead, the System Leads and the Delivery Lead; (3) supervisor approval for changes affecting a DEPI milestone; and (4) an update to the requirements specification and this plan with a version increment.

1.2.3 Schedule (Gantt Chart in Tabular Form)

The schedule below runs from project initiation to final defence. Dates are working days within the stated range. Dependency notation: FS = Finish-to-Start; SS = Start-to-Start.

ID	Task / Phase	Start	End	Dependencies	Lead Assignee
T0	Project Initiation: team formation, roles, repository and tooling setup	1 Oct 2026	5 Oct 2026	n/a	Asmaa Hany
T1	Requirements Gathering: stakeholder analysis, stories, use cases, FR/NFR specification (v1.0)	1 Oct 2026	9 Oct 2026	T0 (SS)	Asmaa Hany
T2	Project Planning & Management (this chapter)	5 Oct 2026	16 Oct 2026	T1 (FS for scope baseline)	Asmaa Hany

ID	Task / Phase	Start	End	Dependencies	Lead Assignee
T3	Literature Review	5 Oct 2026	16 Oct 2026	T1 (SS)	Asmaa Hany
T4	System Analysis & Design: architecture, ERD, API contracts, UML	12 Oct 2026	6 Nov 2026	T1 (FS)	Asmaa Hany, Saad Walid
T5	UX & Discovery lead : wireframes, design system, high-fidelity prototypes	12 Oct 2026	6 Nov 2026	T1 (FS)	Hager Saad
T6	Sprint 0 (Foundation): environments, repository, CI/CD skeleton, container baseline, DB migration framework, design tokens	12 Oct 2026	23 Oct 2026	T0 (FS)	Yara Mokhtar, Manar Gomaa
T7	Sprint 1: AUTH and RBAC (Epic 1); Property & Unit listing core and public search (Epic 2); audit log	26 Oct 2026	6 Nov 2026	T4 (SS), T5 (SS), T6 (FS)	Ahmed Fouda, Hager Saad
T8	Design Sign-off & Architecture Freeze (milestone M2)	6 Nov 2026	6 Nov 2026	T4, T5 (FS)	Asmaa Hany
T9	Sprint 2: Lease & documents with e-signature (Epic 3); Financial Engine, invoicing, payments, ledger (Epic 4); Maintenance & work orders (Epic 5)	9 Nov 2026	20 Nov 2026	T7, T8 (FS)	Saad Walid, Manar Gomaa, Ahmed Fouda
T10	Third-party integration hardening: gateway webhooks, e-signature callbacks, notification providers, object storage	9 Nov 2026	25 Nov 2026	T7 (FS); runs alongside T9	Manar Gomaa
T11	Sprint 3: Communication hub (Epic 6); Reporting, dashboards, exports (Epic 7); cross-epic integration; Should-Have completion	23 Nov 2026	30 Nov 2026	T9 (FS)	Ahmed Fouda, Hager Saad
T12	Continuous QA: unit and integration tests, E2E suites, accessibility audits	26 Oct 2026	2 Dec 2026	T7 (SS)	Yara Mokhtar
T13	Code Freeze & Implementation Delivery (milestone M3)	30 Nov 2026	30 Nov 2026	T11 (FS)	Ahmed Fouda
T14	Sprint 4 (Stabilisation): performance test, security scan, penetration test (structured), bug fixing, regression	1 Dec 2026	3 Dec 2026	T13 (FS)	Yara Mokhtar, Saad Walid
T15	Final Documentation & Test Report: traceability matrix, KPI evidence, deployment and developer guides	23 Nov 2026	3 Dec 2026	T12 (SS), T13	Yara Mokhtar
T16	Final Presentation, Demo & Defence (milestone M4)	4 Dec 2026	4 Dec 2026	T14, T15 (FS)	All; coordinated by Yara Mokhtar

Critical path: T1 → T4 → T8 → T9 → T11 → T13 → T14 → T16. Slippage in design sign-off (T8) or in Sprint 2 (T9) directly threatens the 30 Nov implementation milestone.

Sprint Plan Summary.

Sprint	Dates	Goal	Primary Epics / Requirements
Sprint 0	12 to 23 Oct	Engineering foundation	CI/CD, environments, DB migrations, design tokens
Sprint 1	26 Oct to 6 Nov	Identity and listings	AUTH (FR-AUTH-01 to -13), PROP (FR-PROP-01 to -08, -14)
Sprint 2	9 to 20 Nov	Revenue and operations	LEASE, FIN, MAINT Must-Haves; webhook idempotency and ledger
Sprint 3	23 to 30 Nov	Communication, reporting, integration	COMM, RPT, remaining Should-Haves; code freeze
Sprint 4	1 to 3 Dec	Stabilise and defend	Testing, performance and security validation, documentation, demo rehearsal

1.2.4 Resource Allocation

Human Resources

Six team members with defined roles (see Section 1.3). Each member is allocated to a primary lane with a secondary backup to avoid single points of failure.

Development Environments

Environment	Purpose	Notes
Local	Developer machines using containers (Docker Compose)	Reproducible setup documented in the developer guide (NFR-MNT-06).
CI	Automated build, test, static analysis and dependency scans on every commit	Required by NFR-MNT-05 and NFR-SEC-09.
Staging	Production-equivalent environment for integration, performance and UAT	Used for demos and sprint reviews; connected to sandboxes only.
Production-like demo	Stable snapshot for the final presentation	Seeded with realistic data; protected by a code freeze.

Cloud Infrastructure (provider-agnostic, e.g., AWS)

Resource	Intended Use	Related Requirement
Container hosting (e.g., ECS/EKS or equivalent)	Stateless, horizontally scalable application tier	NFR-PERF-09, NFR-MNT-09
Managed relational database (e.g., PostgreSQL)	Transactional store, ACID ledger, point-in-time recovery	NFR-REL-03, NFR-REL-07
Object storage (e.g., S3) with CDN	Property photos, maintenance media, signed documents; signed expiring URLs	NFR-SEC-12, NFR-PERF-10
Queue / background workers	Invoicing, reminders, notifications, media processing	NFR-PERF-11
Secrets manager and KMS	Credential storage, key management	NFR-SEC-02, NFR-SEC-16
Monitoring and logging stack	Metrics, centralised logs, tracing, alerts	NFR-REL-09

Cost control: free tiers and student credits, small instance sizes, scheduled shutdown of non-production environments, and billing alerts.

Third-Party Staging Sandboxes

Service	Account Type	Purpose
Payment gateway (e.g., Stripe)	Test mode with test keys, test cards and webhook CLI/endpoints	Online payment, tokenisation, refunds, webhook and idempotency testing (Epic FIN).
E-signature provider (e.g., DocuSign or Dropbox Sign)	Developer/sandbox account	Envelope creation, signing callbacks, audit certificates (Epic LEASE).
Email provider (e.g., SES, SMTP sandbox)	Sandbox / verified test domain	Transactional email (Epic COMM).
SMS provider	Trial account with verified test numbers	Emergency and payment-failure alerts.
Mapping/geocoding provider	Free tier	Map-based search (FR-PROP-07, Should-Have).

All integrations are implemented behind provider-agnostic interfaces (NFR-MNT-12) so that sandbox providers can be swapped without redesign.

Collaboration Tooling

Git hosting with protected branches and pull-request review, task board, Figma for design, Postman/Swagger UI for API contracts, a shared documentation repository, and a team chat channel for daily communication.

1.3 Task Assignment & RACI Matrix

1.3.1 Roles & Responsibilities Overview

Team Member	Role	Ownership Areas	Key Responsibilities
Ahmed Fouda	No- Code Application Lead	Frontend and client application; product backlog; application-level integration	Acts as Product Owner; translates stories into frontend features; leads implementation of tenant, manager and owner portals; coordinates frontend-backend contracts; prioritises the backlog; chairs sprint reviews.
Asmaa Hany	No- Code System Lead (documentation and analysis authority)	First four DEPI items: Project Planning & Management, Literature Review, Requirements Gathering, System Analysis & Design; final documentation	Authors and maintains the requirements baseline, project plan, risk register and KPIs; leads system analysis and design (UML, ERD, architecture views, traceability matrix); owns change control; ensures every requirement is traceable to design, test and delivery; coordinates the final report.
Saad Walid	No- Code System Lead (backend and architecture)	Backend APIs, domain model, Financial Engine, database	Designs and implements REST APIs (OpenAPI), RBAC and data-scope enforcement, the double-entry ledger, invoicing scheduler and late-fee engine; leads architecture implementation and database migrations.
Yara Mokhtar	Delivery Lead	Delivery process, CI/CD, QA and release	Facilitates Scrum ceremonies; owns the CI/CD pipeline and environments; leads QA strategy, automated testing, performance and security testing, release readiness and the test report; tracks delivery KPIs.
Manar Gomaa	Integration Lead	Third-party integrations and background processing	Implements and tests gateway integration (idempotency, signature-verified webhooks, reconciliation), e-signature integration (callbacks, retries), notification providers and object storage; owns resilience patterns (timeouts, retries, circuit breakers).
Hager Saad	UX& Discovery Lead	User experience, design system, accessibility	Produces wireframes, prototypes and the shared design system; defines accessible, mobile-first layouts (WCAG 2.1 AA); conducts usability testing; supports frontend implementation of design.

Highlight: Asmaa Hany's contribution and ownership

Asmaa Hany carries ownership of the foundational engineering and documentation phases on which all later work depends:

1. **Project Planning & Management:** this chapter, including scope baseline, schedule, RACI, risk register and KPI framework.
2. **Literature Review:** the research justification and comparative analysis for the platform and its design choices.
3. **Requirements Gathering:** the 28-page specification v1.0 (stakeholder analysis, 11 user stories with acceptance criteria, 3 detailed use cases, 108 functional requirements, NFRs with measurable targets).
4. **System Analysis & Design:** architecture, data model, interface and behaviour modelling that govern implementation.
5. **Traceability and change control:** the link between requirements, design, tests and delivery, and the final documentation package for the 4 Dec presentation.

1.3.2 RACI Matrix

Legend: R = Responsible (does the work); A = Accountable (single owner who signs off); C = Consulted (two-way input); I = Informed (one-way updates). "A/R" indicates the same person is both accountable and responsible. Every deliverable has exactly one A.

Deliverable	Ahmed Fouda (Application Lead)	Asmaa Hany (System Lead)	Saad Walid (System Lead)	Yara Mokhtar (Delivery Lead)	Manar Gomaa (Integration Lead)	Hager Saad (UI/UX Lead)
Project Planning & Management (incl. risk and KPI framework)	C	A/R	C	R	I	I
Literature Review	C	A/R	R	I	R	C
Requirements Specification	C	A/R	R	C	C	C
Architectural Design (architecture, ERD, API contracts, UML)	C	A/R	R	I	C	C
UI/UX Prototypes	C	C	I	I	I	A/R
Frontend & Mobile Client	A/R	I	C	I	C	R
Backend APIs & Financial Engine	R	C	A/R	I	C	I
Third-Party Integrations (payment, e-signature, notifications, storage)	C	C	R	C	A/R	I
CI/CD Pipeline	C	I	C	A/R	R	I
QA & Automated Testing	R	C	R	A/R	R	C
Final Documentation & Presentation	R	A/R	R	R	R	R

Reading the matrix. Asmaa Hany is accountable for every foundational artefact (planning, literature, requirements, design) and for the final documentation package. Backend, frontend, integration, CI/CD and design deliverables each have a single accountable lead from the corresponding role, which prevents ambiguity in sign-off.

1.3.3 Epic Ownership Map

Epic	Primary Owner	Supporting
Epic 1: AUTH	Saad Walid	Ahmed Fouda (UI flows), Hager Saad (UX)
Epic 2: PROP	Ahmed Fouda	Saad Walid (API), Hager Saad (listing UX)
Epic 3: LEASE	Asmaa Hany	Saad Walid (lease model), Ahmed Fouda (UI)
Epic 4: FIN	Hager Saad	Manar Gomaa (gateway), Yara Mokhtar (financial test suite)
Epic 5: MAINT	Yara Mokhtar	Manar Gomaa (media storage), Saad Walid (SLA engine)
Epic 6: COMM	Manar Gomaa	Ahmed Fouda (inbox UI), Hager Saad (Discovery lead)
Epic 7: RPT	Saad Walid	Asmaa Hany (dashboards), Ahmed Fouda (visualization)

1.4 Risk Assessment & Mitigation Plan

1.4.1 Risk Assessment Approach

Risks are identified from the architecture (integrations, financial integrity, file handling), the schedule (fixed DEPI deadlines, small team) and the requirements baseline. Each risk is rated for Probability (P) and Impact (I) as Low, Medium or High. Priority is determined by combining both (H/H = critical). The register is reviewed at every sprint retrospective by the Delivery Lead and Asmaa Hany, and new risks are logged as they emerge.

1.4.2 Risk Register

ID	Category	Risk Description	P	I	Proactive Mitigation & Contingency Strategy	Owner
R-01	External Integration	**Payment webhook idempotency failure.** Duplicate or replayed gateway webhooks, or retried payment requests, cause double postings, duplicate charges or ledger inconsistency.	M	H	**Mitigation:** idempotency keys on all payment requests tied to the invoice ID (FR-FIN-07); webhook signature verification, 5-minute timestamp tolerance and event-ID deduplication (FR-FIN-08, NFR-SEC-08); unique database constraints on event IDs; ACID transactions. **Testing:** replay and out-of-order webhook tests; gateway test cards for failures. **Contingency:** daily reconciliation job; ledger corrections only by compensating entries; manual reconciliation runbook.	Manar Gomaa
R-02	Technical	**Financial ledger inconsistency.** Rounding errors, partial or split payments, or concurrent updates produce ledger imbalances.	M	H	**Mitigation:** append-only double-entry ledger (FR-FIN-14); debit = credit invariant enforced in the database; money stored in minor units with defined rounding rules; at least 90% unit-test coverage for the financial module (NFR-MNT-04); property-based tests. **Contingency:** freeze postings, run reconciliation, and issue reversing entries with audit trail.	Saad Walid
R-03	External Integration	**E-signature provider downtime or callback dropouts.** Signed documents or completion callbacks are lost, leaving leases in "Pending Signature".	M	M	**Mitigation:** provider-agnostic interface (NFR-MNT-12); exponential back-off retries; callback endpoint with idempotent handling plus periodic polling of envelope status; signature expiry and reminder rules (FR-LEASE-06). **Contingency:** documented manual upload of a wet-signed copy with justification (UC-03 flow E2); sandbox mock server for demonstration continuity.	Manar Gomaa
R-04	Technical	**File storage throughput bottlenecks.** High-resolution photos and 50 MB videos from mobile networks slow uploads and thumbnailing, or exhaust storage.	M	M	**Mitigation:** resumable chunked uploads (NFR-PERF-10); direct-to-object-storage uploads using pre-signed URLs; asynchronous thumbnail and malware-scan workers; client-side image compression; CDN delivery; file-size and count limits (FR-MAINT-02). **Contingency:** queue backlog alerts, temporary stricter limits, lifecycle rules to archive old media.	Manar Gomaa
R-05	Operational	**Scope creep around accounting and reporting.** Stakeholders request ledger features, tax rules, ERP sync or custom reports beyond the baseline.	H	M	**Mitigation:** Won't-Have items frozen (FR-FIN-19, FR-RPT-14); MoSCoW discipline; formal change-control procedure; CSV export as the integration path (FR-FIN-18, Could-Have); fixed report catalogue (FR-RPT-01 to -13). **Contingency:** log new requests to a "future release" backlog; trade-off against equal-effort Should-Haves.	Asmaa Hany
R-06	Timeline / Resource	**Ambitious scope versus a seven-week build window.** 63 Must-Have and 27 Should-Have requirements must be delivered by 30 Nov with a six-person team.	H	H	**Mitigation:** Must-Haves built first; vertical-slice delivery per sprint; early foundation in Sprint 0; story-point estimates and velocity monitoring from Sprint 1; reuse of framework features and component libraries. **Contingency:** re-prioritise lowest-value Should-Haves via change control; defer Could-Haves; use feature flags to hide incomplete work.	Yara Mokhtar

ID	Category	Risk Description	P	I	Proactive Mitigation & Contingency Strategy	Owner
R-07	Timeline / Resource	**Fixed DEPI deadlines overlapping with documentation load.** Design, implementation and report writing run in parallel.	H	M	**Mitigation:** dual-track planning; documentation owner (Asmaa Hany) separate from code leads; documentation tasks scheduled in the sprint backlog; early drafting and template reuse. **Contingency:** shift review effort within the team; supervisor consulted early on slippage.	Asmaa Hany
R-08	Timeline / Resource	Team member unavailability or knowledge silos (illness, exams, workload).	M	H	**Mitigation:** primary and backup owner per epic; pair programming and code review; shared documentation; ADRs (NFR-MNT-06). **Contingency:** reassign stories per RACI; reduce Sprint scope.	Yara Mokhtar
R-09	Technical	**Authorisation defects (broken access control / IDOR).** Users access other tenants' or owners' data.	M	H	**Mitigation:** server-side RBAC with object-level ownership checks on every endpoint (FR-AUTH-06/07, NFR-SEC-06); authorisation test coverage across all endpoints; code review checklist. **Contingency:** immediate patch, session revocation, audit-log review and breach-response procedure (NFR-SEC-17).	Saad Walid
R-10	Technical	**Performance degradation at billing-day peaks.** Invoice runs and payments overload the system (target 2,000 concurrent users, 300 req/s bursts).	M	M	**Mitigation:** stateless horizontally scalable application tier (NFR-PERF-09); queue-based workers (NFR-PERF-11); database indexing and slow-query logging (NFR-PERF-08); load tests in Sprint 4. **Contingency:** scale out workers, cache read-heavy queries, stagger invoice batches.	Yara Mokhtar
R-11	External Integration	Notification provider failures or rate limits (email deliverability, SMS cost and opt-in rules).	M	L	**Mitigation:** queued delivery with retries and exponential back-off (FR-COMM-11); fall back to in-app notification; opt-in checks (FR-COMM-05). **Contingency:** switch provider through the abstraction layer; critical notices still surface in dashboards.	Manar Gomaa
R-12	External Integration	Sandbox limitations or API changes by third parties (differences between sandbox and production, deprecations, expired trial accounts).	M	M	**Mitigation:** early spike integrations in Sprint 0/1; pinned SDK and API versions; contract tests; mock servers for offline development. **Contingency:** fall back to mocks for demonstration; alternative provider through the interface.	Manar Gomaa
R-13	Technical	Security vulnerabilities in code or dependencies (OWASP Top 10, vulnerable libraries, leaked secrets).	M	H	**Mitigation:** static and dependency scanning in CI (NFR-SEC-09); secrets manager and no secrets in source control (NFR-SEC-16); structured penetration test before launch (NFR-SEC-10). **Contingency:** critical findings remediated within 7 days; release blocked until resolved.	Yara Mokhtar
R-14	Operational	Data loss or environment failure during development or the demo (database corruption, cloud outage).	L	H	**Mitigation:** automated daily backups and point-in-time recovery (NFR-REL-03); infrastructure as code; seeded demo data scripts. **Contingency:** restore from backup; local containerised fallback demo environment.	Yara Mokhtar

ID	Category	Risk Description	P	I	Proactive Mitigation & Contingency Strategy	Owner
R-15	Operational	Usability and accessibility shortfalls (WCAG AA gaps, mobile-first issues) discovered late.	M	M	**Mitigation:** shared design system; automated accessibility checks (axe, Lighthouse) in CI; early usability testing with at least 8 participants (NFR-USA-04). **Contingency:** prioritise fixes on tenant core flows (pay rent, submit request).	Hager Saad
R-16	Operational	Regulatory and privacy non-compliance (GDPR/local data-protection requirements, consent, retention).	L	M	**Mitigation:** data minimisation, configurable retention, consent capture, data-subject access and erasure support (NFR-SEC-13/14); synthetic data in all non-production environments. **Contingency:** compliance review checklist before demo; remove optional processing.	Asmaa Hany

1.4.3 Risk Heat Map Summary

Impact → / Probability ↓	Low	Medium	High
High	n/a	R-05, R-07	**R-06 (critical)**
Medium	R-11	R-03, R-04, R-10, R-12, R-15	R-01, R-02, R-08, R-09, R-13
Low	n/a	R-16	R-14

Placement uses Probability (rows) and Impact (columns). Critical items (high impact with medium or high probability) are reviewed at every sprint planning and stand-up.

1.4.4 Risk Management Process

1. **Identify:** at sprint planning and retrospectives, and when design decisions are made.
2. **Assess:** assign P and I ratings; record in the register.
3. **Respond:** accept, reduce, transfer (e.g., use a PCI-DSS Level 1 gateway to transfer card-data risk), or avoid (e.g., excluding Won't-Have scope).
4. **Monitor:** risk owners report status weekly; triggers (e.g., velocity below plan for two consecutive sprints) escalate to the Delivery Lead and supervisor.
5. **Review:** closed or new risks are updated in the register at each milestone.

1.5 Key Performance Indicators (KPIs)

KPI targets for technical metrics are directly derived from the non-functional requirements. Business and operational targets are measured during user acceptance testing and demonstration on seeded data, and would be re-measured in a live pilot.

1.5.1 Technical Performance Metrics

KPI ID	Metric	Target	Source Requirement	Measurement Method
KPI-T1	Page load: Largest Contentful Paint (dashboard and listing pages, simulated 4G)	≤ 2.5 s	NFR-PERF-01	Lighthouse / WebPageTest
KPI-T2	API response time p95 / p99 (excluding third-party calls)	p95 < 500 ms; p99 ≤ 1.5 s	NFR-PERF-02	Load testing (e.g., k6, JMeter)
KPI-T3	Search and filter latency over 100,000 listings	≤ 2 s at p95	NFR-PERF-03	Load test on seeded dataset
KPI-T4	Payment initiation processing time (excluding gateway)	≤ 3 s at p95	NFR-PERF-06	APM traces
KPI-T5	System availability	≥ 99.9% monthly uptime	NFR-REL-01	Uptime monitoring, health checks
KPI-T6	Ledger integrity: unexplained variance in daily gateway-ledger reconciliation	Zero variances	NFR-REL-07	Automated reconciliation report
KPI-T7	Unit-test coverage (business logic / financial module)	≥ 80% / ≥ 90%	NFR-MNT-04	Coverage tool in CI
KPI-T8	Unresolved critical or high vulnerabilities at release	0	NFR-SEC-09	SAST, dependency and container scans
KPI-T9	Accessibility score (Lighthouse) and WCAG conformance	≥ 90; WCAG 2.1 AA	NFR-USA-02	Lighthouse, axe, manual audit
KPI-T10	Authorisation test coverage	100% of endpoints	NFR-SEC-06	Automated authorisation test suite
KPI-T11	Recovery objectives (RPO / RTO)	≤ 15 min / ≤ 4 h	NFR-REL-04	Restore drill
KPI-T12	Deployment duration with rollback	< 30 min, one-click rollback	NFR-MNT-05	CI/CD pipeline timings

1.5.2 Project Execution KPIs

KPI ID	Metric	Definition	Target
KPI-P1	Sprint velocity	Story points completed per sprint; stability tracked as the variance from the rolling average.	Established in Sprint 1; variance within ±20% thereafter.
KPI-P2	On-time milestone completion	Number of DEPI milestones delivered on or before the deadline (16 Oct, 6 Nov, 30 Nov, 4 Dec 2026).	4 of 4 (100%)
KPI-P3	Scope delivery	Requirements delivered versus committed (burn-up chart).	100% of 63 Must-Haves; ≥ 80% of 27 Should-Haves
KPI-P4	Defect density	Open critical defects at release; defects per KLOC tracked as a secondary measure.	< 1 critical defect at release (NFR-REL-10)
KPI-P5	MTTR	Mean time to restore service after a critical incident in staging/demo environments.	< 1 hour (NFR-REL-10)
KPI-P6	Requirements traceability	Share of requirements mapped to design elements, code modules and test cases.	100% of Must-Haves; ≥ 90% of Should-Haves
KPI-P7	Code review and CI health	Pull requests merged with at least one review and a passing pipeline.	100%
KPI-P8	Risk exposure trend	Count of High-priority open risks per sprint.	Non-increasing trend after Sprint 2.

1.5.3 Business & Operational KPIs

KPI ID	Metric	Definition	Target (UAT / Pilot)
KPI-B1	User adoption rate	Share of invited users (tenants, managers, owners, contractors) who activate their account and complete at least one core task.	≥ 80% of pilot users
KPI-B2	Rent payment self-service conversion	Share of issued invoices paid online through the portal (versus recorded offline by a manager).	≥ 85% of invoices
KPI-B3	Autopay enrolment	Share of active tenants enrolled in autopay.	≥ 30% of pilot tenants
KPI-B4	Invoice automation rate	Invoices generated automatically without manual intervention.	100%
KPI-B5	Maintenance ticket resolution turnaround	Average time from submission to verified completion, by priority; SLA compliance rate.	SLA compliance ≥ 90%; baseline SLA targets (configurable, NFR-MNT-07): Emergency response 1 h, High 4 h, Normal 24 h, Low 72 h.
KPI-B6	Task efficiency	Pay rent in at most 4 clicks from login; submit a maintenance request in under 3 minutes.	Met by ≥ 85% of test participants (NFR-USA-04/05)
KPI-B7	Usability score	System Usability Scale (SUS).	≥ 75 (NFR-USA-05)
KPI-B8	Owner reporting timeliness	Time to generate a 12-month owner statement or rent roll for 1,000 units.	≤ 10 s (NFR-PERF-07)

1.5.4 KPI Governance

Aspect	Approach
Data collection	CI pipeline reports (coverage, scans), load-test and Lighthouse outputs, task-board metrics, usability test sessions.
Review cadence	Technical KPIs at each sprint review; execution KPIs weekly; business KPIs during UAT in Sprint 3 and Sprint 4.
Ownership	Technical KPIs: Yara Mokhtar and Saad Walid. Execution KPIs: Yara Mokhtar. Business/usability KPIs: Hager Saad and Ahmed Fouda. Consolidated KPI reporting: Asmaa Hany.
Reporting	KPI evidence is compiled into the final Test & Reports package for the 4 Dec 2026 presentation.

Appendix A: Traceability of Chapter 1 to Requirements Baseline

Chapter 1 Element	Requirements Baseline Reference
Objectives OBJ-1 to OBJ-5	Epics AUTH, PROP, LEASE, FIN, MAINT, COMM, RPT; NFR categories
Scope table	Requirements Summary (63/27/11/7 = 108)
Out-of-scope list	FR-AUTH-15, FR-PROP-15, FR-LEASE-15, FR-FIN-19, FR-MAINT-17, FR-COMM-13, FR-RPT-14
Risk register	UC-02 (E2, E5), UC-03 (E2), UC-01 (E3), NFR-SEC, NFR-REL, NFR-PERF
KPIs	NFR-PERF-01 to -11, NFR-SEC-06/09, NFR-REL-01/04/07/10, NFR-USA-02/04/05, NFR-MNT-04/05

End of Chapter 1